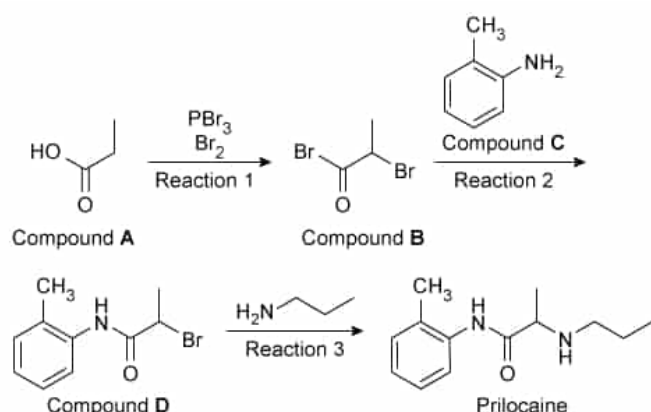


Local anesthetics are a class of drugs that cause the cessation of pain and sensation at or near the point of administration. In contrast, the use of general anesthetics leads to the onset of unconsciousness. Local anesthetics typically exert their pharmacological action through the blockade of voltage-gated sodium channels present on peripheral nerves. After a local anesthetic has been administered, sodium ions are unable to rapidly enter the nerve cell to initiate an action potential in response to external stimuli that otherwise would induce a pain response.

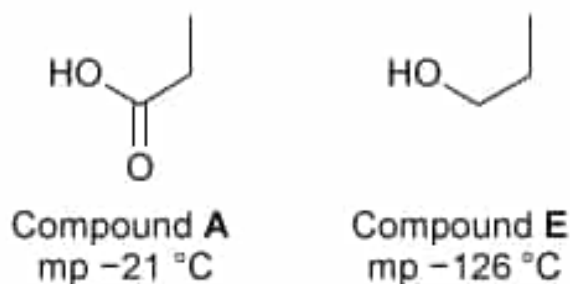
The natural product cocaine was one of the first compounds identified with potent local anesthetic activity. Given the issues associated with cocaine use, several synthetic local anesthetics have been developed, including procaine (Novocain), benzocaine, lidocaine, and prilocaine.

Prilocaine is commonly employed in dentistry, either as a standalone agent or in combination with other local anesthetics. Prilocaine is commercially prepared from Compound **A** through the three-step synthesis depicted in Scheme 1.



Scheme 1

Which statement correctly describes the basis for the observed melting point difference between Compound **A** and Compound **E**?

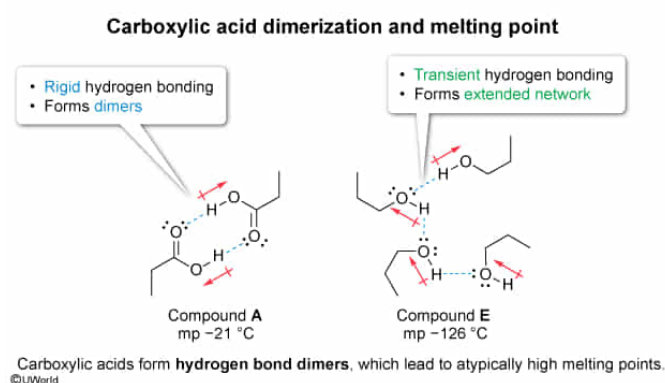


- ☐ A. Compound **E** forms dimers, which increase its melting point. [8%]
- ☐ B. Compound **A** forms dimers, which decrease its melting point. [8%]
- ☐ C. Compound **E** forms dimers, which decrease its melting point. [8%]
- ✔ ☒ D. Compound **A** forms dimers, which increase its melting point. [74%]

Correct

75% answered correctly

Explanation:



The **melting point** of a substance is the temperature at which it undergoes a transition between the solid phase and liquid phase. As a sample is heated, the kinetic energy of individual molecules increases. When the kinetic energy exceeds the energy of attractive **intermolecular forces** in a solid, individual molecules are no longer rigidly held in a solid-state lattice and the molecule undergoes a **phase transition** to a liquid.

Molecules with a greater amount of attractive **intermolecular forces** have higher melting points. Since the amount of **London dispersion forces** is proportional to the electrons in a molecule, London dispersion forces are larger for molecules with higher molecular weight. As a result, given similar functional groups, melting points will generally increase with increasing molecular weight.

As a type of **dipole–dipole** interaction, **hydrogen bonding** between molecules leads to a dramatic increase in melting point. **Carboxylic acids** have **even higher melting points** due to their ability to form **hydrogen bond dimers**—a phenomenon that effectively *doubles* the molecular weight of the molecules entering the liquid state.

In this question, the melting point of Compound **A** ($-21\text{ }^{\circ}\text{C}$) is much **higher** than the melting point of Compound **E** ($-126\text{ }^{\circ}\text{C}$), despite a difference in molecular weight of only 14 g/mol . Compound **A** is a **carboxylic acid that forms hydrogen bond dimers** between molecules, whereas Compound **E** is an alcohol that forms a network of *transient* hydrogen bonds (*without* rigid dimers). Therefore, **Compound A forms dimers that increase its melting point.**

(Choice A) Compound **E** is an alcohol that is unable to form hydrogen bond dimers. Additionally, the melting point of Compound **E** is *lower* than the melting point of Compound **A**.

(Choice B) Although the melting point of Compound **A** ($-21\text{ }^{\circ}\text{C}$) has a smaller absolute value, it is over $100\text{ }^{\circ}\text{C}$ warmer than the melting point of Compound **E** ($-126\text{ }^{\circ}\text{C}$).

(Choice C) Compound **E** is an alcohol that is unable to form hydrogen bond dimers. Furthermore, it is the *absence* of hydrogen bond dimers that leads to the lower melting point of Compound **E**.

Educational objective:

The melting point of a compound is greatly influenced by the nature of its intermolecular forces. Carboxylic acids form hydrogen bond dimers that greatly increase the compound's melting point.

Time spent:0 sec

Subject:
Organic Chemistry

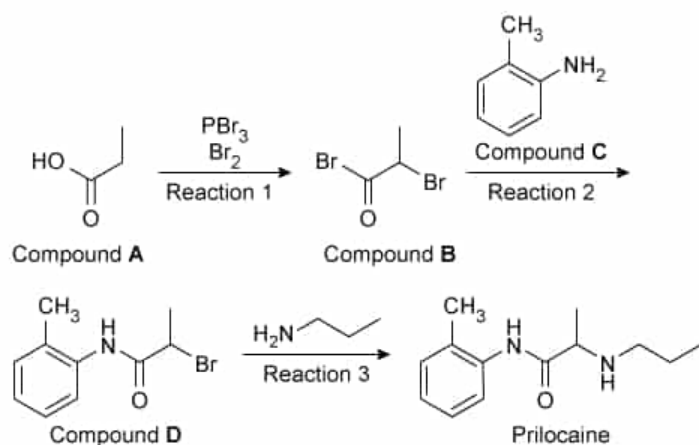
QID:403972

System:
Functional Groups and Their Reactions

Local anesthetics are a class of drugs that cause the cessation of pain and sensation at or near the point of administration. In contrast, the use of general anesthetics leads to the onset of unconsciousness. Local anesthetics typically exert their pharmacological action through the blockade of voltage-gated sodium channels present on peripheral nerves. After a local anesthetic has been administered, sodium ions are unable to rapidly enter the nerve cell to initiate an action potential in response to external stimuli that otherwise would induce a pain response.

The natural product cocaine was one of the first compounds identified with potent local anesthetic activity. Given the issues associated with cocaine use, several synthetic local anesthetics have been developed, including procaine (Novocain), benzocaine, lidocaine, and prilocaine.

Prilocaine is commonly employed in dentistry, either as a standalone agent or in combination with other local anesthetics. Prilocaine is commercially prepared from Compound **A** through the three-step synthesis depicted in Scheme 1.



Scheme 1

The molecular weight of Compound **A** is 74 g/mol, the molecular weight of Compound **C** is 107 g/mol, and the weight of a bromine atom is 80 g/mol. Based on the passage, what is the molecular weight of Compound **D**?

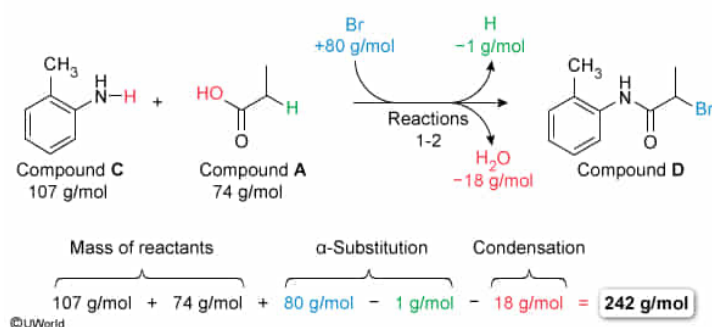
- ☐ A. 163 g/mol [3%]
- ✓ ☒ B. 242 g/mol [27%]
- ☐ C. 243 g/mol [49%]
- ☐ D. 260 g/mol [19%]

Correct

27% answered correctly

Explanation:

Calculating molecular weight for a multi-step synthesis



The **molecular weight (MW)** of a substance can be determined using a variety of approaches:

- With a **molecular formula**, MW can be calculated using the stoichiometry of elements and each element's respective atomic weight.
- **Structural formulas** provide an alternative method of determining molecular formula and therefore molecular weight.
- MW is always an integer multiple of the molecular weight for an **empirical formula**.
- **Mass spectrometry** provides a **molecular ion** that has a m/z value equal to its MW.

The information in a synthetic transformation can also be utilized to calculate the molecular weight of a component substance. When a given reaction or reaction sequence is described in terms of the **net changes** taking place, logic can be applied to deduce the desired MW.

Based on the passage, the net conversion of Compounds **A** and **C** to

Compound **D** takes place over two steps (Reactions 1 and 2). Through this net process:

- The **amine** in Compound **C** (107 g/mol) is **condensed** with the **carboxylic acid** in Compound **A** (74 g/mol), **eliminating a molecule of water** (−18 g/mol).
- An α-hydrogen in Compound **A** is lost (−1 g/mol) and replaced by an α-bromo group (+80 g/mol).

The sum of these MW changes (107 g/mol + 74 g/mol + 80 g/mol − 1 g/mol − 18 g/mol) leads to the conclusion that the MW of Compound **D** is 242 g/mol.

(Choice A) A MW of 163 g/mol is calculated when only the condensation component is considered and the α-bromination is ignored (107 g/mol + 74 g/mol − 18 g/mol).

(Choice C) A MW of 243 g/mol is calculated when both the condensation and α-bromination components are considered, but the loss of the α-hydrogen is ignored (107 g/mol + 74 g/mol + 80 g/mol − 18 g/mol).

(Choice D) A MW of 260 g/mol is calculated when both the condensation and α-bromination components are considered, but the loss of a molecule of water is ignored (107 g/mol + 74 g/mol + 80 g/mol − 1 g/mol).

Educational objective:

The molecular weight of a substance can be determined through several processes. Molecular weights of intermediates in a multi-step synthesis can be calculated by identifying the net reaction(s) taking place.

Time spent: 0 sec

Subject:
Organic Chemistry

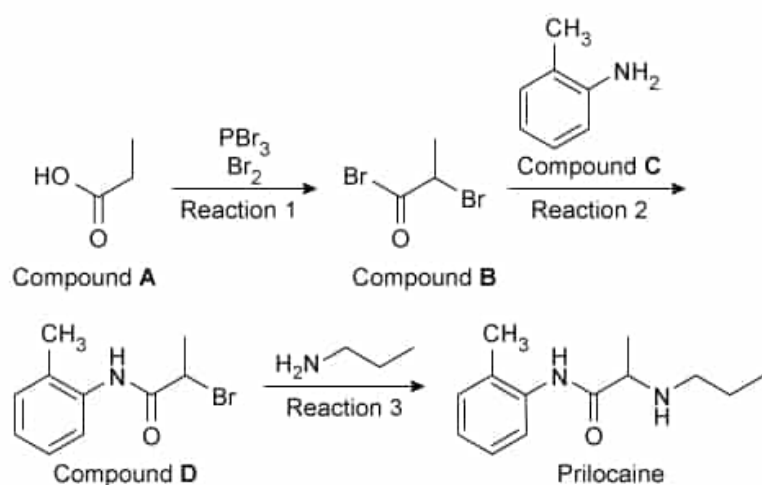
QID: 403973

System:
Functional Groups and Their Reactions

Local anesthetics are a class of drugs that cause the cessation of pain and sensation at or near the point of administration. In contrast, the use of general anesthetics leads to the onset of unconsciousness. Local anesthetics typically exert their pharmacological action through the blockade of voltage-gated sodium channels present on peripheral nerves. After a local anesthetic has been administered, sodium ions are unable to rapidly enter the nerve cell to initiate an action potential in response to external stimuli that otherwise would induce a pain response.

The natural product cocaine was one of the first compounds identified with potent local anesthetic activity. Given the issues associated with cocaine use, several synthetic local anesthetics have been developed, including procaine (Novocain), benzocaine, lidocaine, and prilocaine.

Prilocaine is commonly employed in dentistry, either as a standalone agent or in combination with other local anesthetics. Prilocaine is commercially prepared from Compound **A** through the three-step synthesis depicted in Scheme 1.



Scheme 1

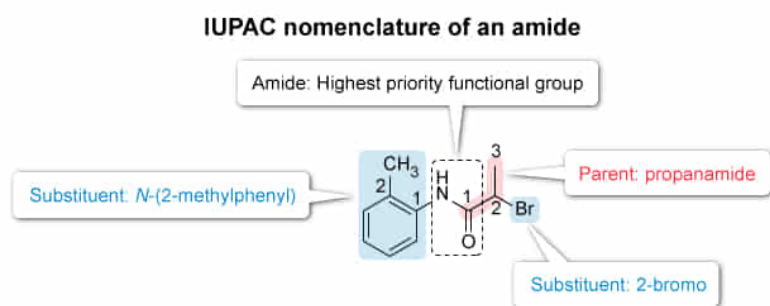
According to the IUPAC, what is the systematic name for Compound **D**?

- ☐ A. 2-methyl-*N*-(2-bromopropanoyl)aniline [8%]
☐ B. 2-bromo-1-(2-methylphenyl)propanamide [17%]
☐ C. 1-(1-bromoethyl)-*N*-(2-methylphenyl)methanamide [9%]
✔ ☒ D. 2-bromo-*N*-(2-methylphenyl)propanamide [64%]

Correct

64% answered correctly

Explanation:



The IUPAC name for Compound **D** is **2-bromo-*N*-(2-methylphenyl)propanamide**
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IUPAC provides a systematic method of naming organic compounds. A compound's **parent chain** is the longest continuous chain of carbon atoms that contains the greatest number of the **highest-priority functional group** present, and it is numbered to provide the highest-priority functional group(s) the lowest possible number. The parent chain name includes a **prefix** denoting the number of carbon atoms in the chain, a middle portion denoting the presence or absence of select carbon-carbon pi bonds, and a suffix denoting the quantity and type of highest-priority functional group.

Lower-priority functional groups are named as **substituents** that include a **numerical position of attachment** to the parent chain.

When a substituent's point of attachment is a **heteroatom** (ie, oxygen, nitrogen, sulfur), the point of attachment is denoted by an **italicized capital letter** (ie, **O-**, **N-**, **S-**). Substituent chains are numbered from the point of attachment to the parent chain and named using a prefix to indicate the carbon chain length and a *-yl* suffix.

The highest-priority functional group in Compound **D** is the **amide**, a **carboxylic acid derivative**. There are **three carbons** in the longest continuous chain and the **parent chain will be propanamide**. One substituent (bromo) is located at carbon 2 (ie, **2-bromo-**), while the other complex substituent is bonded to the amide nitrogen and is given a location of **N-**. The aromatic **phenyl ring** is numbered from its point of attachment to the parent chain and contains a methyl group at its position 2 (ie, **-N-(2-methylphenyl)-**).

Consequently, the IUPAC name for Compound **D** is **2-bromo-N-(2-methylphenyl)propanamide**.

(Choice A) The parent chain must include the highest priority functional group. 2-Methyl-N-(2-bromopropanoyl)aniline incorrectly identifies **aniline** (an aromatic amine) as the parent, when a higher-priority amide is present.

(Choice B) Substituents attached to the nitrogen of an amide are not given a numerical position—they are denoted with a location of **N-**. 2-Bromo-1-(2-methylphenyl)propanamide incorrectly identifies the 2-methylphenyl substituent as being bonded to carbon 1.

(Choice C) The parent chain of an amide includes the longest continuous chain of carbons. 1-(1-Bromoethyl)-N-(2-methylphenyl)methanamide incorrectly limits the parent chain to one carbon (methanamide), when the two carbons in 1-(1-

bromoethyl) should also be included.

Educational objective:

Amides are a high-priority functional group that are named using the suffix *–amide*. Substituents bonded to the nitrogen atom of an amide are given a location of *N-*, rather than a numerical point of attachment.

Time spent:0 sec

Subject:
Organic Chemistry

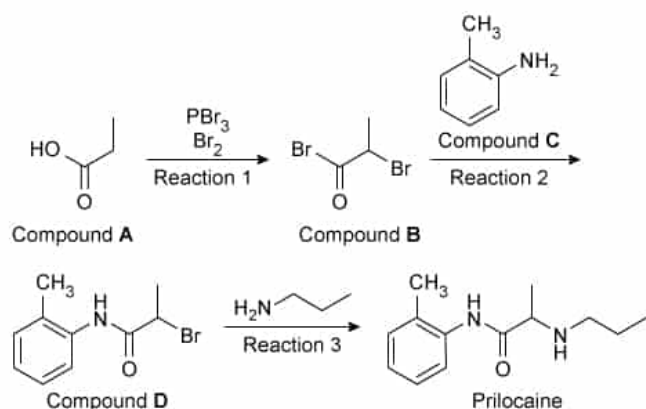
QID:403974

System:
Functional Groups and Their Reactions

Local anesthetics are a class of drugs that cause the cessation of pain and sensation at or near the point of administration. In contrast, the use of general anesthetics leads to the onset of unconsciousness. Local anesthetics typically exert their pharmacological action through the blockade of voltage-gated sodium channels present on peripheral nerves. After a local anesthetic has been administered, sodium ions are unable to rapidly enter the nerve cell to initiate an action potential in response to external stimuli that otherwise would induce a pain response.

The natural product cocaine was one of the first compounds identified with potent local anesthetic activity. Given the issues associated with cocaine use, several synthetic local anesthetics have been developed, including procaine (Novocain), benzocaine, lidocaine, and prilocaine.

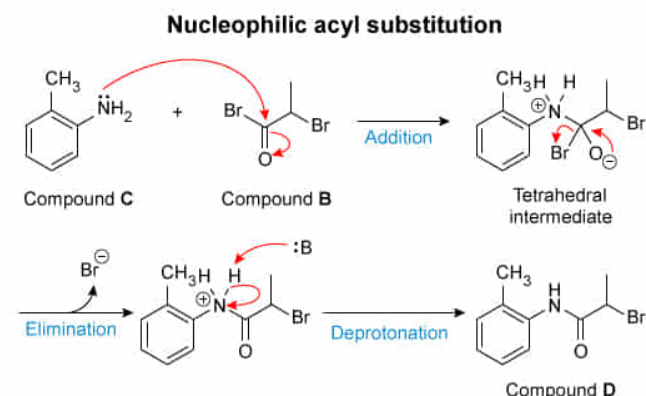
Prilocaine is commonly employed in dentistry, either as a standalone agent or in combination with other local anesthetics. Prilocaine is commercially prepared from Compound **A** through the three-step synthesis depicted in Scheme 1.



Scheme 1

The transformation in Reaction 2 is best categorized as a:

- ☐ A. nucleophilic addition. [28%]
- ☐ B. nucleophilic elimination. [4%]
- ✓ ☒ C. nucleophilic acyl substitution. [49%]
- ☐ D. $\text{S}_{\text{N}}2$ reaction. [18%]

Explanation:

A nucleophilic acyl substitution reaction follows an addition–elimination mechanism and proceeds through a tetrahedral intermediate.

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The **mechanism** of a chemical reaction is the step-by-step sequence of **elementary steps** that describe changes taking place at the molecular level. Reaction mechanisms with two or more elementary steps are called **multi-step reactions** and typically generate temporary reaction **intermediates** that are later consumed en route to the final product. **Combining** the elementary steps for a reaction mechanism generates the balanced net reaction for the transformation.

Organic chemistry uses broad classifications to describe reaction mechanisms. **Substitution reactions** involve the displacement of one group by another. **Addition reactions** involve two groups adding across a pi bond, consuming the pi bond in the process. **Elimination reactions** involve the loss of two groups from adjacent carbon atoms and the formation of a new pi bond. Consequently, addition and elimination reactions are **reciprocal processes**. Furthermore, many reaction mechanisms are described as either **nucleophilic** or **electrophilic**, depending on the reactant's role at key points in the reaction mechanism.

Reaction 2 depicts the conversion of Compounds **B** (a **carboxylic acid halide**) and **C** (an **aniline**) to Compound **D** (an **amide**). The net displacement of a bromine atom in Compound **B** with the nitrogen of Compound **C** qualifies Reaction 2 as a **substitution reaction**.

This substitution takes place on the carbonyl carbon in Compound **B** (an **acyl**

carbon) and does not take place in a single elementary step. Instead, the **nucleophilic** Compound **C** first attacks the electrophilic carbon in Compound **B**, generating a **tetrahedral intermediate**, which collapses to reform the pi bond and eliminate the bromine as a **leaving group**.

Therefore, the best categorization of Reaction 2 is a **nucleophilic acyl substitution**.

(Choice A) Although the first elementary step of Reaction 2 can be classified as a **nucleophilic addition** due to consumption of the pi bond, the tetrahedral intermediate collapses, *reforming* the pi bond and resulting in *no net consumption* of the pi bond by the end of the net reaction.

(Choice B) The net process for Reaction 2 involves the loss of only *one* group and a new pi bond does *not* form.

(Choice D) **S_N2 reactions** involve a nucleophile displacing a leaving group in a single elementary step. Reaction 2 has a *two-step* mechanism.

Educational objective:

A reaction mechanism describes the sequence of elementary steps for a given chemical transformation. Organic mechanisms can be broadly classified based on their features. Nucleophilic acyl substitutions involve the displacement of one group for another on a carbonyl carbon.

Time spent:0 sec

Subject:
Organic Chemistry

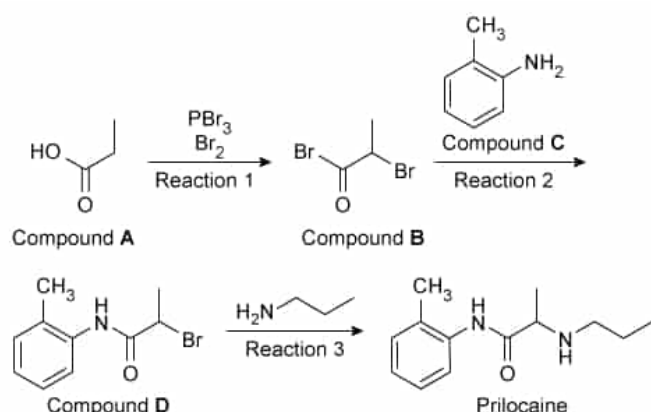
QID:403975

System:
Functional Groups and Their Reactions

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Prilocaine is commonly employed in dentistry, either as a standalone agent or in combination with other local anesthetics. Prilocaine is commercially prepared from Compound **A** through the three-step synthesis depicted in Scheme 1.



Scheme 1

Which statement best explains why Reaction 2 proceeds predominantly in the forward direction as written?

- ☐ A. The carbonyl carbon in Compound **B** is less electrophilic than the carbonyl carbon in Compound **D**. [10%]
- ☒ B. The carbonyl carbon in Compound **B** is more electrophilic than the carbonyl carbon in Compound **D**. [52%]
- ☐ C. The carbonyl carbon in Compound **B** is more sterically hindered

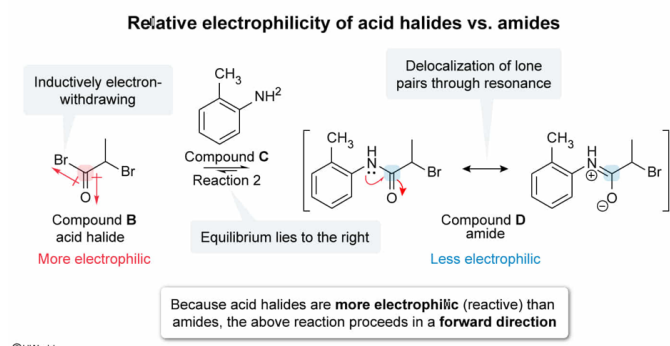
than the carbonyl carbon in Compound **D**. [3%]

- ☐ D. The carbonyl carbon in Compound **B** is less sterically hindered than the carbonyl carbon in Compound **D**. [33%]

Correct

53% answered correctly

Explanation:



The **principle of microscopic reversibility** states that a process that can proceed in a forward direction has a corresponding process that can proceed in the reverse direction. An implication of this principle is that all chemical reactions should be viewed as **dynamic equilibria** between forward and reverse processes. Often, when one component of a dynamic equilibrium is significantly disfavored, the dynamic nature of the process is simplified to a unidirectional process; in these cases, an equilibrium arrow is replaced by a standard arrow.

Steric effects and **electronic effects** represent two major ways by which reaction outcomes are justified. Steric effects generally relate to the **size** or **proximity** of functional groups near a reaction center, where **larger groups** tend to decrease the rate of reaction. In contrast, electronic effects encompass molecular features involving electron density, including **polarity**, **resonance**, or **conjugation**. The major players in electronic effects are electrons outside the sigma bond network: **lone pairs** or **pi bonds**.

Carboxylic acid derivatives differ in the group bonded to the carboxylic acid carbonyl and are interconverted through **nucleophilic acyl substitution reactions**. The relative reactivity of carboxylic acid derivatives results from a combination of steric effects and electronic effects.

This question asks about the molecular basis for the preferential conversion of Compound **B** (an **acid halide**) to Compound **D** (an **amide**). Acid halides

are the most reactive carboxylic acid derivatives due to the presence of the **electron-withdrawing** halogen atom that **inductively** removes electron density away from the carbonyl carbon atom, **greatly increasing its electrophilicity**. In contrast, amides are far less reactive (ie, less electrophilic) due to the resonance-delocalization of the nitrogen lone pair into the carbonyl.

Because reactions tend to proceed from more reactive to less reactive components, the forward process in Reaction 2 is favored due to the **greater electrophilicity of Compound B over Compound D**.

(Choice A) The electronegative bromine atom in Compound **B** inductively removes electron density from its carbonyl carbon and increases its rate of conversion to Compound **D**. As a result, Compound **B** is a *stronger* electrophile than Compound **D**.

(Choices C and D) The relative reactivity of Compounds **B** and **D** results from *electronic effects* of the electronegative bromine atom, rather than *steric effects*.

Educational objective:

Many organic reactions take place due to a combination of steric effects and electronic effects. The chemical reactivity of carboxylic acid derivatives arises primarily due to electronic effects, rather than steric effects.

Time spent:0 sec

Subject:
Organic Chemistry

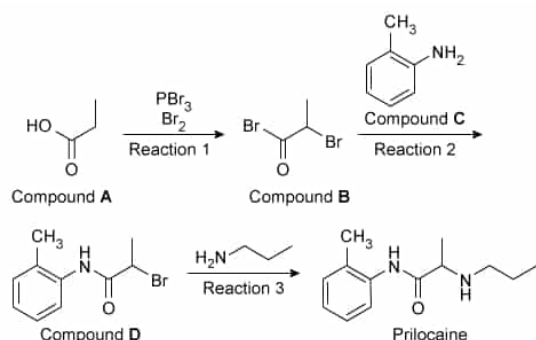
QID:403976

System:
Functional Groups and Their Reactions

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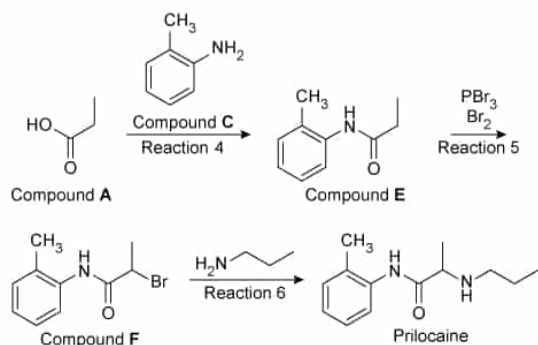
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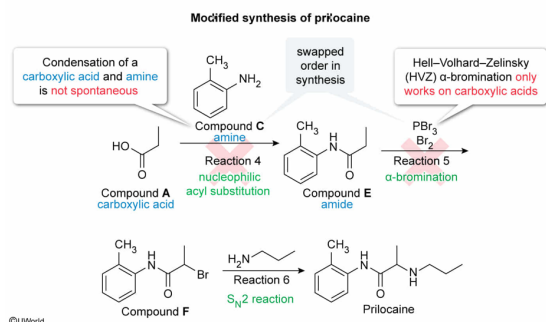


Scheme 1

The synthesis below is a modified version of the synthesis in the passage. Is the modified synthesis likely to be successful in the preparation of prilocaine?



- ☐ A. Yes, because the outcome of prilocaine will still be achieved. [6%]
- ✓ ☒ B. No, because Reaction 4 creates the first synthetic roadblock. [61%]
- ☐ C. No, because Reaction 5 creates the first synthetic roadblock. [29%]
- ☐ D. No, because Reaction 6 creates the first synthetic roadblock. [2%]

Explanation:

In **multi-step organic synthesis**, two or more chemical transformations are successively performed toward a final synthetic goal. Some multi-step syntheses require discrete synthetic steps to take place in a specific order while others are capable of greater sequence flexibility.

Carboxylic acid derivatives differ in the group bonded to the carboxylic acid carbonyl and are interconverted through **nucleophilic acyl substitution reactions**. The relative reactivity of carboxylic acid derivatives results from a combination of steric effects and electronic effects.

In the modified synthesis presented in this question, the sequence of the first two reactions is **swapped**, resulting in the nucleophilic acyl substitution (Reaction 4) now occurring before α -bromination (Reaction 5). Consequently, the **nucleophilic acyl substitution** step occurs between Compound A (a **carboxylic acid**) and Compound C (an **amine**). The condensation of a carboxylic acid and an amine to an **amide** (Compound E) is **not spontaneous**, as the carboxylic acid lacks a good **leaving group** and is less electrophilic than the original **acid halide**. Rather, Reaction 4 would promote a **neutralization reaction** between the acid and base. For these reasons, **Reaction 4 is the earliest step in the modified synthesis** that presents roadblocks toward the production of prilocaine.

(Choice A) Both Reactions 4 and 5 present roadblocks in the modified synthesis, likely *preventing* the successful production of prilocaine through this approach.

(Choice C) By shifting the placement of the **Hell-Volhard-Zelinsky (HVZ) bromination**, Reaction 5 now uses an amide substrate rather than a carboxylic acid. While the HVZ reaction is specific for carboxylic acids, this roadblock comes *after* the earlier roadblock at Reaction 4.

(Choice D) Reaction 6 has the same relative placement, conditions, and substrate as the original synthesis. Consequently, it would *not* be expected to present a roadblock in the modified synthesis.

Educational objective:

Discrete steps in multi-step organic syntheses may either provide opportunity for sequence flexibility or need to occur in a specific sequence. The condensation reaction between a carboxylic acid and an amine is not spontaneous and typically requires additional earlier steps to activate the carboxylic acid for nucleophilic attack.

Time spent: 0 sec

Subject:

Organic Chemistry

QID: 403977

System:

Functional Groups and Their Reactions